



24-Acre Regenerative Agriculture Research & Soil Systems Study

 **Soil Seed & Water | University of Arizona Collaboration**

A Five-Year Field Research Program



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This research program evaluates multiple soil-building strategies, biological amendments, and crop rotations to determine whether a scalable, profitable regenerative farming system can be developed for the desert Southwest.

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Desert Southwest Regenerative Agriculture

Building resilient soil systems for arid environments

1. EXECUTIVE SUMMARY

This five-year field research program evaluates multiple soil-building strategies, biological amendments, and crop rotations to determine whether a **scalable, profitable regenerative farming system** can be developed for the desert Southwest.

★ KEY PROGRAM ELEMENTS

- ✓ Industrial hemp grown annually from January–June as the primary economic crop
- ✓ Rotations of grains and legumes to build soil fertility and support biological succession
- ✓ Six soil profile treatments evaluated across randomized research plots
- ✓ University of Arizona collaboration for rigorous scientific validation

Study Overview

24-Acre Field Layout



 Block D subdivided into 100' × 100' plots for replicated university trials

Soil Treatments Under Evaluation

Treatment	Description	Base Application
1. Green Waste Compost	High-carbon municipal green waste material	10 tons/acre
2. Dairy Compost	Nitrogen-rich dairy manure compost	10 tons/acre
3. Chicken/Dairy Blend	10% chicken + 90% dairy compost mixture	10 tons/acre
4. Desert Resilience Blend	Biochar, zeolite, humic acids, worm castings, mineral fines	10 tons/acre
5. Dairy + Green Waste	Balanced blend of dairy and green waste composts	10 tons/acre
6. Compost + Fungal Extract	Standard compost enhanced with fungal inoculum	10 tons/acre

© **Final Deliverable:** A validated regenerative crop rotation system for desert soils, including soil blend rankings, economic comparisons, and a farmer-ready Transition Guide.

2. PROBLEM STATEMENT / RATIONALE

Agriculture in the desert Southwest faces worsening challenges that threaten long-term farm viability and environmental sustainability:



Environmental Pressures

- Depleted soil carbon levels
- Severe compaction issues
- Salinity accumulation
- Rising temperatures and heat stress
- Water scarcity and drought



Economic Pressures

- Rising fertilizer costs
- Increased water expenses
- Heavy equipment capital requirements
- Reduced long-term yields
- Soil degradation costs

 **The Industrial Farming Problem:** Current industrial systems rely heavily on tillage, chemical fertilizers, herbicides, and large machinery. These practices increase costs while degrading soil biology, creating long-term economic and environmental vulnerabilities.

The Regenerative Alternative

Regenerative systems offer a promising alternative built upon:



Compost &
Biological Inputs



Mycorrhizal
Networks



Water-Efficient
Soil Structures

Research Needs Addressed

This study addresses the critical need for region-specific data by providing:

- ✓ **Clear comparisons** between soil blends and treatments
- ✓ **Understanding of crop adaptability** in desert conditions
- ✓ **Insight into biological succession** from bacterial to fungal-dominant soils
- ✓ **Realistic input cost modeling** for farmer decision-making
- ✓ **Transitional pathway** from industrial to regenerative agriculture

Goal: Identify what works *and* what does not, ensuring farmers avoid unproductive soil systems, unsuitable crops, or economically unfavorable strategies.

3. METHODS

3.1 Study Site

Location Details

- **Total Area:** 24 acres
- **Environment:** Irrigated desert
- **Configuration:** Four 6-acre blocks
- **Research Plots:** 100' × 100' subdivisions

Study Design

- Randomized complete block design
- Replicated trials within UA plots
- Annual data collection cycles
- 5-year longitudinal tracking

3.2 Soil Treatments

All treatments receive **10 tons compost/acre + 1 ton gypsum/acre** in Year 1. Client may choose 4 or all 6 treatments.

1

Green Waste Compost

High-carbon municipal green waste. Supports fungal colonization and long-term carbon building.

2

Dairy Compost

Nitrogen-rich with strong bacterial activity. Excellent for rapid nutrient cycling.

3

Chicken/Dairy Blend

10% chicken + 90% dairy. Higher phosphorus and balanced nutrition profile.

4

Desert Resilience Blend

Biochar, zeolite, humic acids, worm castings, mineral fines. Optimized for arid conditions.

5

Dairy + Green Waste

Balanced blend combining nitrogen availability with carbon stability.

6

Compost + Fungal Extract

Standard compost enhanced with fungal inoculum for accelerated succession.

 **Optional Control:** A bare-soil control plot may be included for baseline comparison against all treatments.

 3.3 CROP ROTATIONS

Three crop rotations per year support biological succession from **bacterial** → **balanced** → **fungal-leaning** soils:

 **Hemp (Jan–Jun)**

 **Summer (Jul–Sep)**

 **Fall (Oct–Dec)**

 **Rotation 1: Industrial Hemp (Primary Crop) — January to June**

Grown annually as the primary economic crop. Provides exceptional root density, canopy cover, and market value.

Benefits: Deep taproot system, high biomass production, economic output, excellent soil structure building

 **Rotation 2: Summer Crops — July to September**

Grains and legumes selected for high biomass production and mycorrhizal compatibility:

Grains	Legumes
- Sorghum - Millet - Teff - Sudangrass	- Cowpeas - Pigeon Pea - Sunhemp

 **Rotation 3: Fall Crops — October to December**

Cool-season grains and legumes for continued soil building:

Grains	Legumes
- Oats - Barley - Rye - Triticale	- Hairy vetch - Crimson clover - Fava bean

📌 **Succession Goal:** This rotation sequence accelerates the transition from bacterial-dominant to fungal-dominant soil biology, building water-holding capacity and long-term fertility.

 4. BIOLOGICAL INPUTS

 4.1 Compost Application Taper (5-Year Schedule)

Year 1 10 tons/acre	Year 2 8 tons/acre	Year 3 6 tons/acre	Year 4 4-5 tons/acre	Year 5 3-4 tons/acre
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 Compost rates decrease as soil biology becomes self-sustaining

 4.2 Gypsum Application Schedule

Year	Application Rate	Notes
Year 1	1 ton/acre	Full application for initial soil conditioning
Year 2	0.5–1 ton/acre	Adjusted based on soil test results
Year 3	0.25–0.5 ton/acre	Maintenance application
Years 4–5	As needed	Only if SAR or infiltration requires correction

 4.3 Fungal Extract Application Protocol

 Application Timing

Applied at **all seasonal transitions** throughout the year:

 Pre-Hemp (January)	 Pre-Summer (July)	 Pre-Fall (October)
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Application Rate: 10-20 gallons/acre per application
Method: Banded drench over crop rows for maximum root zone penetration

5. SOIL MONITORING & MEASUREMENTS

Comprehensive soil monitoring at baseline and annually tracks the progression of soil health across all treatments.



Physical & Biological Metrics

SOC	Soil Organic Carbon
 F:B Ratio	Fungal to Bacterial Ratio
 Bulk Density	Soil compaction measure
 Infiltration	Water absorption rate
 Aggregates	Aggregate Stability



Chemical Metrics

 EC	Electrical Conductivity
 SAR	Sodium Adsorption Ratio
 CEC	Cation Exchange Capacity
 pH	Soil acidity/alkalinity
 NPK	Available N, P, K
 POXC	Active Carbon



Measurement Schedule



Baseline Testing

Complete soil profile analysis before any treatments are applied (Year 0)



Annual Testing

All metrics measured at the same time each year for consistent comparison



Soil Profile Metrics Table

Year 1 through Year 5 progression tracking

\$ 6. ECONOMIC EVALUATION

The study provides comprehensive cost-benefit analysis comparing regenerative systems versus industrial agriculture approaches.

Comparison Metrics

**Input Costs**

- Compost & amendments
- Fungal extract applications
- Gypsum applications
- Seed costs

**Operations**

- Irrigation volume
- Labor requirements
- Equipment needs
- Tractor passes

**Returns**

- Crop yields
- Market value
- Long-term savings
- Reduced inputs

Industrial vs. Regenerative Comparison

Factor	Industrial System	Regenerative System
Capital Equipment	High (large tractors, implements)	Reduced footprint
Fertilizer Dependency	High annual cost, increasing	Decreasing over time
Water Usage	Standard irrigation rates	Reduced (improved infiltration)
Soil Biology	Degrading over time	Improving annually
Long-term Trend	Increasing costs, declining yields	Decreasing costs, stable yields

ROI Model Output

The economic evaluation produces a comprehensive ROI model for farmers evaluating a regenerative transition, including:

- ✓ Break-even analysis by treatment type
- ✓ 5-year cost projections
- ✓ Yield comparison data
- ✓ Water savings calculations
- ✓ Labor efficiency metrics

 7. PROJECT OUTCOMES

At the end of five years, the study will deliver comprehensive, actionable results for farmers and researchers:



Validated System

A proven regenerative crop rotation system specifically designed for desert soils



Ranked Treatments

Soil blends ranked by performance, cost-effectiveness, and biological value



Economic Analysis

Detailed economic comparisons to industrial agriculture with ROI projections



Succession Data

Soil carbon and fungal succession trajectories across all treatments



Farmer-Ready Transition Guide

A comprehensive, field-tested guide enabling farmers to implement the validated system on their own operations, including:

- Step-by-step implementation protocols
- Equipment recommendations
- Application timing calendars
- Crop selection guidance
- Troubleshooting guides
- Cost estimation worksheets
- Soil testing protocols
- Success metrics and KPIs



Deliverable Graphics Package



Soil Profile
Illustrations



Concept
Maps



24-Acre
Field Map



Soil Succession
Graphics



Input Curve
Graphics



Rotation
Diagrams

 8. DESERT REGENERATIVE FIELD PROTOCOL

Grains → Legumes → Industrial Hemp (24 Acres) | Year 1: Soil Jumpstart Program

🎯 **Purpose:** Rebuild microbial life in arid soils, improve infiltration, increase root depth, and support high-performing annual crops using on-site extract production (750 gal/hr) and water bubble for deep drenching.

 8.1 Application Rates (Year 1)

 Grains (Oct–Dec)

Extract: 60 gal/ac
Water: 700–1,000 gal/ac
Timing: Planting or 1–3 leaf
Method: Banded drench

 Legumes (Jun–Sep)

Extract: 50–60 gal/ac
Water: 600–800 gal/ac
Timing: In-furrow or V2–V3
Method: Nodulation zone

 Hemp (Jan–Jun)

Extract: 70 gal/ac
Water: 800–1,200 gal/ac
Timing: Emergence/transplant
Boost: +30 gal/ac @ 21–30 days

 8.2 Field Totals for 24 Acres (Year 1)

 Extract Required

Grains (8 ac × 60): **480 gal**
Legumes (8 ac × 60): **480 gal**
Hemp (8 ac × 70): **560 gal**
TOTAL: 1,520 gal

Extractor run time: $1,520 \div 750 = \sim 2$
hours

 Water Required

Average dilution: ~900 gal/ac
24 ac × 900: **~21,600 gal**

Water bubble loads: **9–11 fills**
(depending on tank size)

 Year 2+ Maintenance Rates

Crop	Extract Rate	Notes
Grains	40 gal/ac	Early season only
Legumes	40 gal/ac	In-furrow or early drench
Hemp	50 gal/ac	Optional mid-season 20 gal/ac

✍ **Expected Progress:** By Year 3, soil should show clear biological improvement and reduced need for heavy drenching.

 8.3 APPLICATION METHODS

 Soil Pre-Conditioning (Optional but Ideal)

If soil is hydrophobic, apply a light pre-wet or run a brief irrigation cycle before extract application.

 Drench Guidelines

- ✓ Use **large droplets and low pressure** to avoid evaporation
- ✓ Maintain consistent band width over crop rows
- ✓ For hemp, ensure water reaches **6–12 inches depth**

 Push-Down Water Method (Optional)

For deeper penetration in very dry desert soils:

- Step 1:** Apply 2/3 of total volume with extract

Step 2: Follow with 1/3 clear water immediately behind

 Mixing & Tank Chart

Extract Rate (gal/ac)	Water Volume (gal/ac)	Total Mix (gal/ac)
50	600	650
60	800	860
70	1,000	1,070
30 (top-up)	400	430

 Water Bubble Example (2,000-gal tank)

For a 60 gal/ac rate at 860 total gal/ac → 2,000-gal tank covers **~2.3 acres per fill**

Crop Sequence Benefits

Grains → Legumes → Hemp rotation builds soil fast:

 **Grains:** Early bacterial boost + surface carbon

 **Hemp:** Deep-root action + fungal stimulation

 **Legumes:** Nitrogen fixing + root turnover

Result: Accelerated desert soil recovery and increased water-holding capacity

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9. FUNGAL-DOMINANT EXTRACT FORMULAS

9.1 Understanding Fungal vs. Bacterial Dominance

⚠ Key Insight: Dairy-based vermicompost is **bacterial-heavy by default**. While excellent for annual crops and quick nutrient cycling, it won't achieve the fungal dominance needed for perennial systems or BEAM-level inoculation.

Bacterial Indicators

- High nitrogen content
- Fine texture materials
- Fast breakdown rates
- Dairy/manure based
- Worm-processed materials

Fungal Requirements

- Carbon & lignin-rich feedstocks
- Wood chips & shredded branches
- Partially decomposed bark
- Leaf litter & straw
- Biochar (holds hyphae)

9.2 The SSW Fungal Formula

Compost Blend for Fungal Extraction

50%	Shredded wood chips + leaf litter
25%	Partially composted mulch/wood fines
15%	Vermicompost
5%	Mature dairy compost
5%	Biochar + forest duff

 Build, wet, and let sit 3–6 months for dark, hyphae-loaded fungal material

Extractor Basket Mix

For each extraction batch:

- 8–10 lbs fungal compost blend
- 2–3 lbs vermicompost
- **NO sugar/molasses**
- Minimal agitation (fungal hyphae are fragile)

❌ **Critical Warning:** Molasses will explode bacterial counts and destroy fungal ratios. Use it **ONLY** when making bacterial teas, never for fungal extracts.

 9.3 HEMP-SPECIFIC FORMULATIONS

 Target F:B Ratio for Hemp: 2:1 to 3:1

Hemp benefits from moderate fungal dominance—not as fungal-loving as grapes, but more fungal than turf.

 What Hemp Benefits From

- ✓ Structured soil with good drainage
- ✓ Early root-colonizing mycorrhizae (liquid form in drip system)
- ✓ Strong nitrogen cycling without ammonia spikes
- ✓ Disease suppression
- ✓ Higher terpene/cannabinoid expression (for CBD/CBC types)

 Hemp Extractor Basket Mix

Component	Percentage	Purpose
Fungal mulch compost	35%	Fungal base
Wood chips + leaf litter	25%	Carbon structure
Vermicompost	20%	Microbial diversity
Dairy compost	10%	Nutrient base
Biochar	8%	Hyphae habitat
Forest duff	2%	Fungal inoculum

 Additives for Hemp Extract

 Recommended

- Humic acid — 3 oz
- Fish hydrolysate — 1–1.5 oz
- Kelp meal extract (optional for root vigor)

 Avoid

- Molasses (bacterial bomb)
- High-dose fish (becomes bacterial)
- EM-1, bokashi (bacterial/acidic)

 Application Timing by Hemp Type



Fiber/Grain Hemp

- Early root establishment (0–2 weeks)
- Vegetative push (week 3–5)



CBD/Cannabinoid Hemp

- Root establishment (0–2 weeks)
- Pre-flower boost (week 4–6)
- Optional mid-flower (for terpenes)

10. APPENDIX

10.1 Field Application Checklist

Before Application

- ☐ Extract brewer functioning
- ☐ Dry extract material on site
- ☐ Water bubble filled
- ☐ Soil moisture test completed
- ☐ Nozzle set for coarse droplet

During Application

- ☐ Maintain constant ground speed
- ☐ Watch for runoff (hydrophobicity)
- ☐ Re-wet if needed
- ☐ Monitor extract flow rate

After Application

- ☐ Document gallons per acre applied
- ☐ Note soil penetration depth achieved
- ☐ Record weather conditions (temp, wind, humidity)
- ☐ Schedule next application pass
- ☐ Clean and store equipment properly

10.2 Forest Duff Collection Guide

What is Forest Duff?

The top 1–3 inches of undisturbed forest floor under trees—dark, spongy, earthy material containing:

- Decomposed leaves & twigs
- Fine roots & fungal hyphae
- Native forest microbes
- Mycorrhizal spores
- Microarthropods
- Humic materials

 Recommended Source: Mogollon Rim (Arizona)

High-elevation forest mulch contains true forest-floor fungi, mycorrhizal spores, and native Arizona decomposers adapted to drought. **1–2 totes** will inoculate hundreds of yards of compost.

 10.3 Basalt Red Cinders Usage

Application	Rate	Benefits
Industrial Hemp	5–8% by volume	Structure, drainage, minerals
Tree Crops	5–10% by volume	Long-term structure, microbe habitat

PROPOSAL SUMMARY

Partnership Opportunity

This five-year research program represents a unique opportunity to develop and validate regenerative agriculture practices specifically designed for the desert Southwest.

★ Key Investment Highlights

24

Acres Under
Management

5

Year Research
Program

6

Soil Treatment
Evaluations

UA

University
Partnership

Deliverables Recap

- ✓ **Validated Regenerative System** — Proven crop rotation and soil management protocols
- ✓ **Treatment Rankings** — Data-driven recommendations for soil blend selection
- ✓ **Economic Analysis** — Complete ROI modeling vs. industrial agriculture
- ✓ **Transition Guide** — Farmer-ready implementation manual
- ✓ **Scientific Publication** — University of Arizona research collaboration

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